

2DTS-M2 Motorized Deterministic Transfer System

Fully Automated Precision Assembly Platform
for 2D Materials & Heterostructures



Intelligent Optical Imaging.
Automated Microscopy Control.
Built for Research & Inspection.

The CryoNano 2DTS-M2 is a fully motorized deterministic dry transfer system engineered for ultra-precise assembly of 2D materials, van der Waals heterostructures, and hybrid layered devices.

Designed for advanced quantum materials research and semiconductor prototyping, the platform enables accurate placement of atomically thin materials onto user-defined locations with exceptional repeatability and alignment precision.

The compact architecture supports glove-box integration for processing air-sensitive materials and allows substrate heating during transfer for enhanced process control.

The integrated CN-Transfer Suite software provides real-time visualization, automated positioning, image capture, and guided transfer workflows for deterministic assembly applications.

KEY FEATURES

- **Fully Motorized Deterministic**

Transfer Motorized XY motion, theta rotation, zoom, and focus enable deterministic sub-micron placement for multilayer stacks and twisted bilayers.

- **High-Precision Linear Motion System**

Linear crossed-roller stages with closed-loop servo control and encoder feedback provide high repeatability with minimal drift.

- **Precision Theta Rotation Stage**

Worm-gear-driven theta rotation with crossed-roller bearing architecture and OFHC copper vacuum chuck ensures stable operation.

- **Integrated Temperature-Controlled Stage**

PID-controlled heating stage supports room temperature to 100°C operation with 0.1°C resolution and uniform heating.

- **Advanced Optical System**

Fully apochromatic motorized optics with motorized zoom and focus control support coaxial illumination and APO objective integration.

Notes

- System performance depends on vibration isolation and environmental stability
- Compatible with inert atmosphere glove-box systems
- Additional optical and illumination modules available
- Custom stage travel and automation configurations supported

APPLICATIONS

- Deterministic 2D material transfer
- Twistronics and moiré structure fabrication
- Graphene and TMDC heterostructure assembly
- Quantum device prototyping
- Air-sensitive material processing inside glove boxes
- Semiconductor and nanoelectronics research



2DTS-M2 Motorized Deterministic Transfer System

SPECIFICATIONS

MOTION SYSTEM: MOTORIZED STAGES & CONTROLLER XYZ PRECISION STAGE (FOCUS & PLANAR POSITIONING)

Axes	X, Y, Z
Drive Mechanism	Precision LM screw drives for smooth, backlash-controlled motion
Guide System	Linear rail guides for high stability and load capacity
Travel Range	◦ X-Axis: 25mm ◦ Y-Axis: 25mm ◦ Z-Axis (Stamp): 25mm
Resolution (Full-Step)	1 μ m
Resolution (Microstepping)	Configurable down to 0.6 μ m (with 1/256 microstepping)
Load Capacity (Horizontal)	5 kg

XY θ PRECISION STAGE (PLANAR & ANGULAR ALIGNMENT)

Axes	X, Y, Theta (θ)
Drive Mechanism	Precision LM guides for smooth, backlash-controlled motion and worm gear for θ
Travel Range	◦ X-Axis: 25mm ◦ Y-Axis: 25mm ◦ θ-Axis: 360° continuous
Resolution (Microstepping)	◦ X/Y: Configurable down to 0.1 μm ◦ θ: Configurable down to 0.005°
Load Capacity (Vertical)	5 kg

MOTION CONTROLLER: FLEXIBLE, 4-AXIS CONTROL

The system is powered by a sophisticated 4-axis motion controller, purpose-built to drive the XYZ and Theta stages with precision and reliability. Embracing modern motion control approaches, this controller delivers high speed and accuracy for complex positioning tasks, regardless of the motor technology employed. It features straightforward USB connectivity for immediate setup and is managed by an intuitive graphical software interface. This interface allows for direct, straightforward control and fine-tuning of all four axes, putting complete command of your positioning system at your fingertips. For advanced applications, multi-axis synchronization and the integration of custom control software are seamlessly supported.

MICROSTEPPING MOTOR CONTROLLER

Control Architecture	Advanced 4-axis (or more) microstepping controller
Motor Drive	High-performance digital drives with selectable microstepping resolution (up to 1/256)
Communication Interface	USB 3.0 / Ethernet for high-speed command and data transfer
I/O & Triggering	◦ 4x Opto-isolated Digital Inputs: For limit switches, emergency stop, and external triggers ◦ Input Power 24V DC universal power supply included

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IMAGING SYSTEM

CAMERA	8 MP CMOS MACHINE VISION CAMERA
Sensor	Sony IMX [or equivalent] CMOS, Global Shutter
Resolution	3840 × 2160 (4K UHD)
Frame Rate	45 fps at full resolution
Pixel Size	~2.0 µm x 2.0 µm
Interface	USB 3.0 (with locking connector for reliability)
Trigger	Hardware trigger input for precise synchronization with motion controller

OPTICAL SYSTEM

MOTORIZED ZOOM MICROSCOPE WITH CONTROLLER

Type	Motorized Zoom Microscope with controller
Zoom Ratio	1:10
Optical Magnification	0.3x – 3.0x
Working Distance	9 mm
Optical Design	Apochromatic 20X lens system for minimal chromatic aberration and high contrast
Motorization	Integrated stepper motors for precise, software-controlled zoom and fine focus adjustments
Illumination	Coaxial (epi) light illumination with power supply

TEMPERATURE CONTROLLER

PRECISION TEMPERATURE CONTROLLER

Controller Type	SSD-based (Solid State Drive) precision temperature controller
Temperature Range	Ambient to 200°C
Control Accuracy	±0.5°C
Display	Digital readout of setpoint and actual temperature
Sensor Type	Thermocouple
Heater Stage	Integrated OHFC heating stage compatible with XYθ motion platforms



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CONTROL & ANALYSIS SOFTWARE

MAIN SOFTWARE INTERFACE

Main Interface	Intuitive Graphical User Interface (GUI) for controlling all system functions
Motion Control	Individual axis jogging and absolute/relative positioning
Microscope Integration	Software sliders for motorized zoom and focus
Vision & Analysis	Live video display with real-time 4K resolution
Image Capture	Single snap, time-lapse, video recording
Measurement Tools	On-screen calibration and measurement tools (distances, angles, areas)
Overlay Features	Overlay of scale bars and annotations
Data Export	Export of images and measurement results (CSV, image formats)